**Tutorial – Three-Phase Inverter**

# Description

This tutorial guides you through the implementation of a **Three-phase inverter model** using the **DL 2106T06** module with bidirectional serial communication between a **Simulink-based Host Model** which will be running in the master (PC) and a **Target Model** running on **DL RCPCORE** module. The system implements basic control, monitoring, and fault protection based on real-time analog acquisition and serial communication.

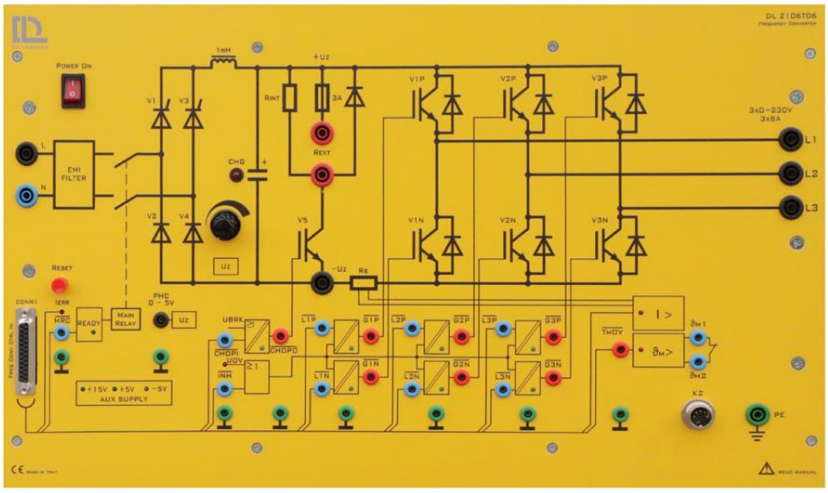
# Learning Objectives

By performing this experiment, students will learn about the **Three-phase inverter**, while reaching the following main objectives:

1. Understand how to configure the DL 2106T06 as a Three-phase inverter.
2. Construct master–target models in Simulink, with the Host model dedicated to debugging and monitoring, while the target model executes control algorithms, performs data acquisition, and integrate protection logic with constraints.
3. Implement serial communication between the host and target.
4. Implement protection logic for safe inverter operation.
5. Monitor DC-link voltage, inverter current, and fault status in real time.

# Part 1: Set up a single-phase inverter

The DL 2106T06 is a module that integrates a controlled single-phase rectifier, a DC link, and three IGBT legs at the output. The module was originally designed to function as a frequency converter. Nevertheless, owing to the independent controllability of its output inverter, it can also be employed to construct other types of power converters. In this tutorial, we use it as a **Three-phase inverter** for an RLC load.



To use the DL 2106T06 frequency converter as a **single-phase inverter**, the system must be reconfigured in the following way:

- **Leg 1** of the output inverter: only the **upper IGBT** (V1P) is controlled using PWM and receives a duty cycle D. The lower IGBT (V1N) receives the complementary duty cycle(1-D).
- **Leg 2**: only the **upper IGBT** (V2P) is controlled using PWM and receives a duty cycle D. The lower IGBT (V2N) receives the complementary duty cycle(1-D).
- **Leg 3**: only the **upper IGBT** (V3P) is controlled using PWM and receives a duty cycle D. The lower IGBT (V3N) receives the complementary duty cycle(1-D).

# Part 2: Construct Simulink Host and Target Models

To open the example for reference, run **openDLDemo('x3p_inv')** in the command window.

Create a folder for the new project

💡 *It is recommended that each project has its own separate folder, and the folder path does not contain any spaces.*

Navigate the MATLAB working directory to the new project folder

Execute the CreateNew command in the MATLAB Command Window and enter the project name in the pop-up dialog box.

💡 *Avoid using special characters and spaces in the project name.*

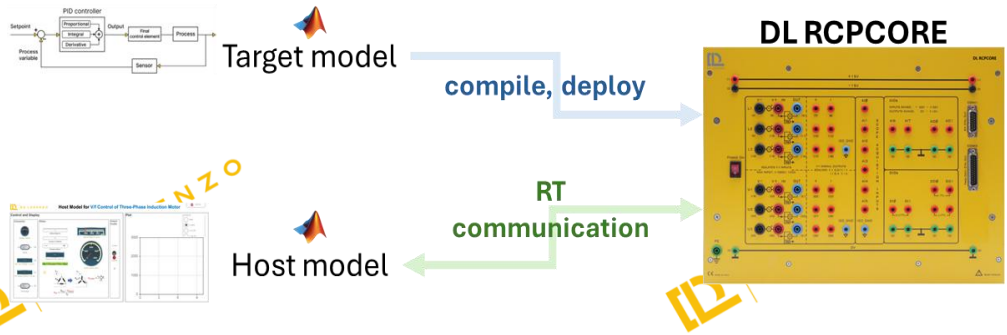
Wait until the "Busy" status in the lower-left corner of MATLAB disappears, and no warnings or errors are displayed in the Command Window.

The user will be asked to name the models created.

Once successfully executed, two models will be created by the system:

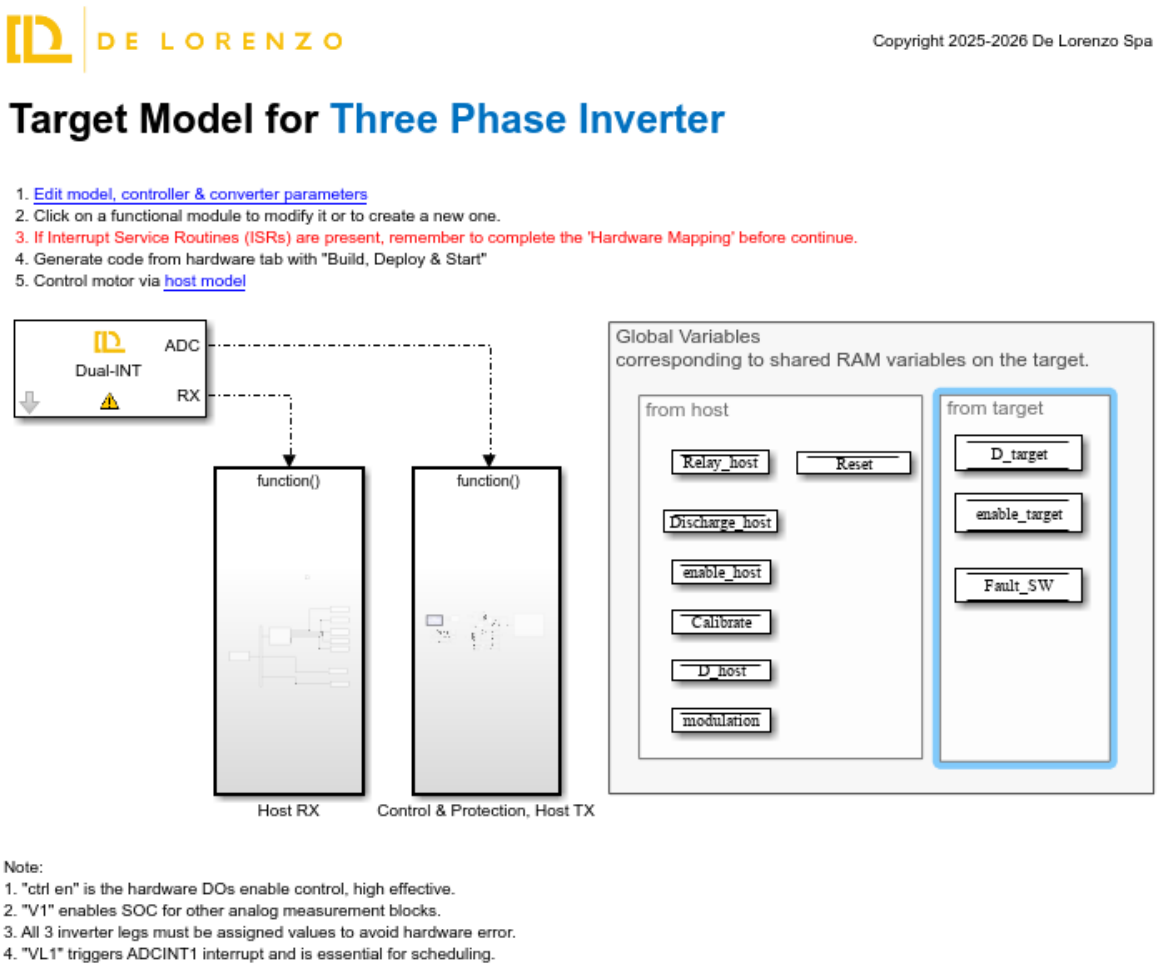
```
Command Window
>> CreateNew
  Model "test_host" created.
  Block "Host Serial Setup" added to model "test_host".
  Configuration applied to model "test_host".
  Script "testparam.m" created.
  Initialization script "testparam.m" set for model "test_host".
  Model "test_target" created.
  Configuration applied to model "test_target".
  Initialization script "testparam.m" set for model "test_target".
  Block "ctrl en" added to model "test_target" and assigned initial value 1.
  Block "V1" added to model "test_target" and assigned initial value 0.
  Block "V2" added to model "test_target" and assigned initial value 0.
  Block "V3" added to model "test_target" and assigned initial value 0.
  Block "VL1" added to model "test_target" and default terminated.
>>
```

- **Target model**: This model will be compiled, deployed, and executed on the DL RCPCORE hardware. Its primary responsibility is to execute control algorithms and support real-time communication with the host model.
- **Host model**: This model runs on the PC in simulation mode. Its role is to debug parameters in the target model and display specific signals.




An **.m file** for setting parameters will also be generated automatically.

## Step 1: Target Model Structure and Basic Blocks




When the *Target Model for the Three-Phase Inverter* is opened, the user is presented with a structured and organized interface that separates the real-time execution logic from the communication layer and shared memory variables.

On the left side, the **Dual-INT block** is visible. This block generates the hardware Interrupt Service Routines (ISRs) required for deterministic real-time operation. It manages both the ADC interrupt (used for periodic control execution) and the RX interrupt (used for serial communication reception).

At the center of the model, two main subsystems are defined:

- **Host RX**: Handles incoming serial communication from the host model. It parses received commands and updates of the corresponding shared RAM variables.
- **Control & Protection, Host TX**: Contains the main real-time control logic, protection algorithms, modulation routines, and the transmission of monitoring signals back to the host.

On the right side, the **Global Variables** section defines the shared RAM variables used for host–target communication. These are divided into:

- **From Host**: Variables transmitted by the host model, such as relay control, discharge command, enable signal, reset command, calibration trigger, duty reference, and modulation selection.
- **From Target**: Variables computed on the target side and transmitted back to the host, including duty command, enable status, and software fault flags.

At the bottom of the interface, important implementation notes are provided. These reminders ensure correct hardware configuration, proper interrupt scheduling, and mandatory inclusion of specific inverter leg and measurement blocks.

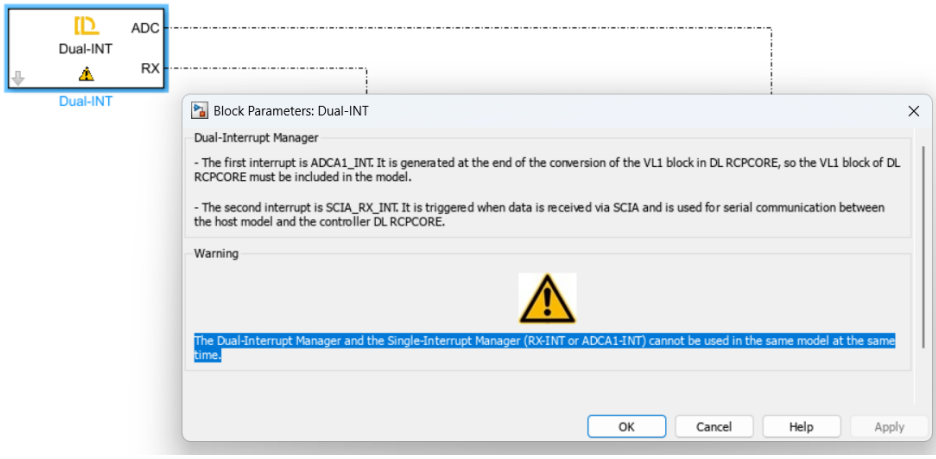
Overall, this interface clearly separates communication, control execution, and shared data handling, providing modular and scalable architecture for inverter implementation.

In this structured model, the essential blocks have been. Read the descriptions and warnings of these blocks to understand their significance and function as essential blocks.

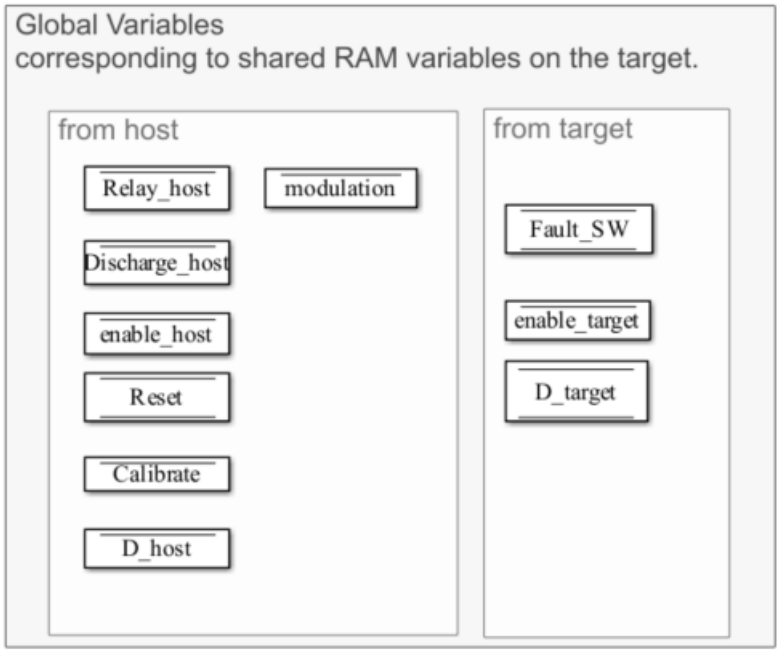
In this section, the users will be guided through a step-by-step procedure to build the model of the three-phase inverter.

Navigate through the Simulink Library Browser to locate the blocks in the DL library that are required to interface the model with the DL PEL-HIL hardware modules. Add the following blocks:

- Dual-INT: The **Dual-Interrupt block** is used to **enable hardware-triggered execution** of two key tasks on the target microcontroller: the **ADC interrupt** (ADC), which ensures precise and timely execution of ADC conversion and related control routines, and the **RX interrupt** (RX), which allows efficient reception of serial data only when needed, saving processing resources. By relying on hardware interrupts instead of fixed sample times, the model becomes more reasonable and practical.

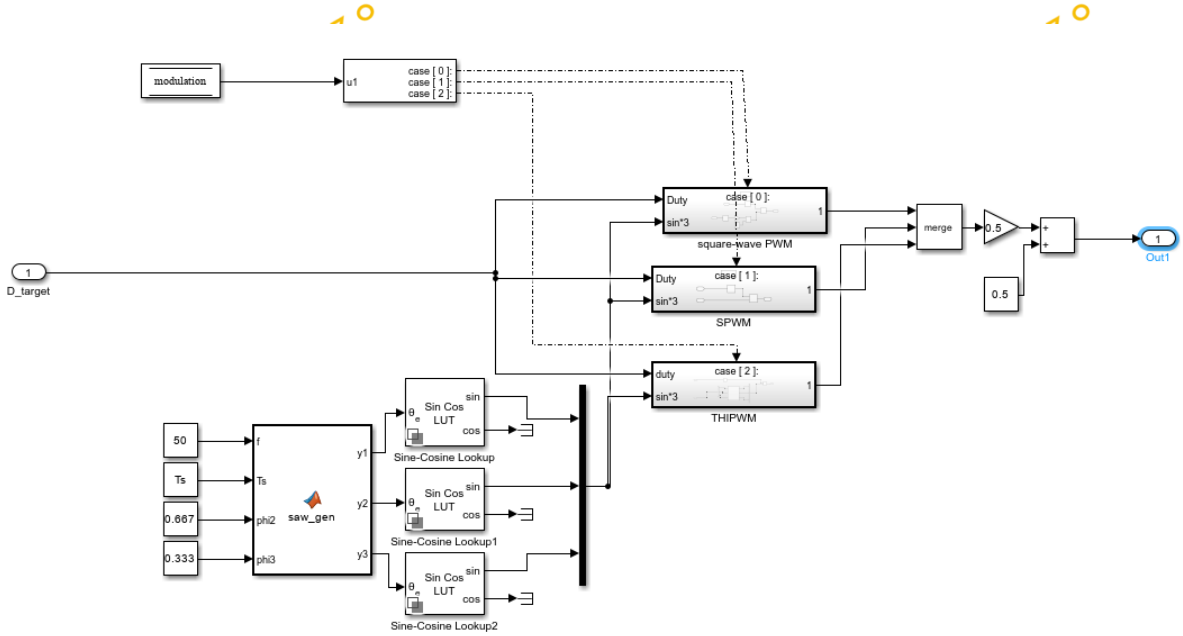



- Preassign key variables in RAM for real-time access using the **Data Store Memory** block.

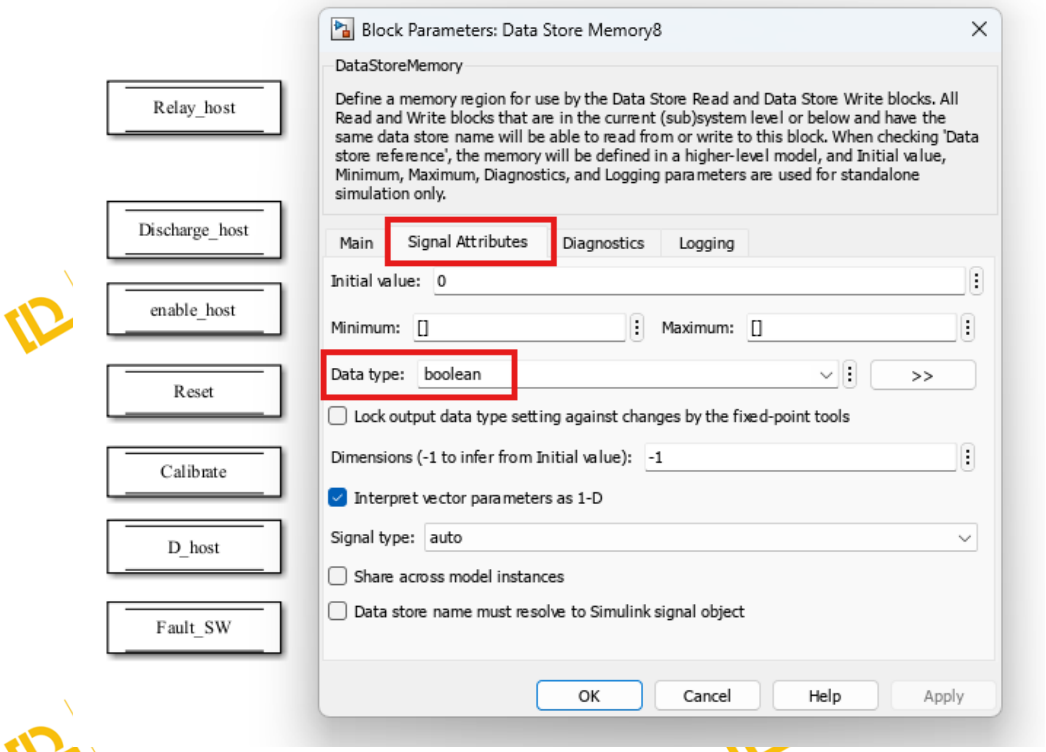



- **Relay_host (Boolean** signal)
  Relay control signal sent from the host; whether the target follows it depends on the protection algorithm.
- **Discharge_host (** Boolean signal)
  DC link discharge control signal sent from the host; whether the target follows it depends on the protection algorithm.
- **enable_host (**Boolean signal)
  Enable control signal sent from the host for IGBT o; whether the target follows it depends on the protection algorithm.
- **Fault_SW (Boolean** signal)
  Based on the results of the software protection algorithm specifically defined for this project. Please distinguish it from hardware fault triggered by hardware protection. The determination of hardware fault is performed entirely by the hardware.
- **Reset (** Boolean signal)
  A signal sent by the host to clear latched software faults.
- **Calibrate (**Boolean signal )
  A signal sent by the host to enable hardware data acquisition offset calibration.
- **D_host (**Single signal)
  A signal sent by the host to drive IGBT V1 with a duty cycle ranging from 0 to 1.
- **Modulation (single signal)**
  A signal sent by the host to activate the chosen modulation technique

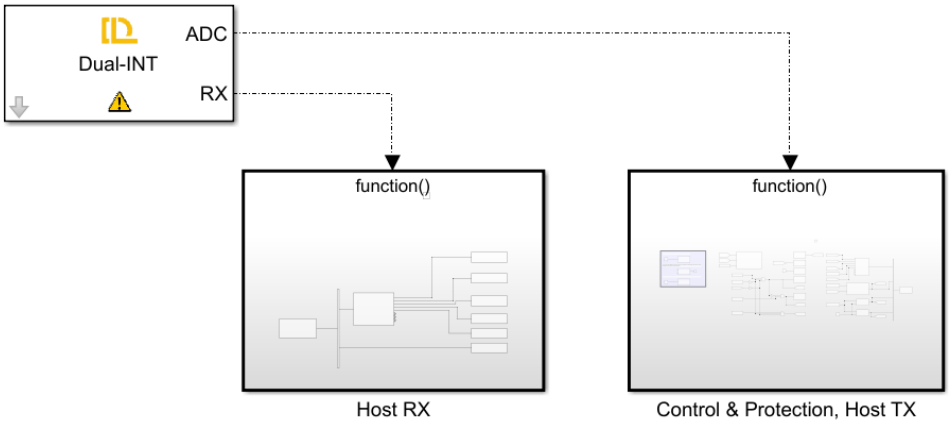
- **0 for square wave modulation.**
- **1 for SPWM modulation.**
- **2 for 3rd harmonic injection modulation.**




💡 *Do not forget to specify the Data type of the stored data through Signal Attributes tab, as it is important for efficient and correct communication. Please refer to* ***Host-Target Communication – Fundamentals in DL PEL-HIL Getting Started to understand the recommended type of data to be used.***

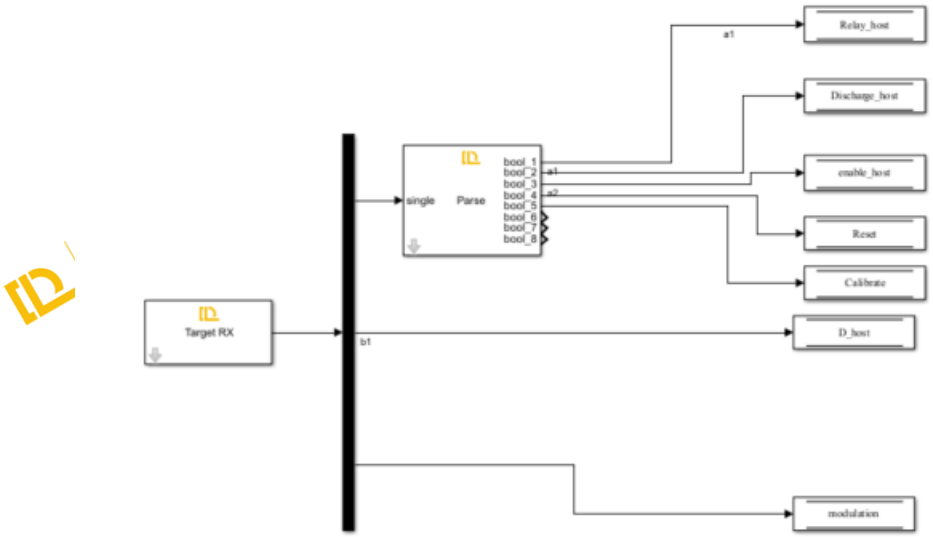



- Create two subsystems and connect them to the two outputs of the Dual-INT block to perform their specified functions as explained before.




**The first subsystem** is for receiving serial data from the host through Target **RX Block** and writing those data to their specified data store memory blocks to allow them to be read elsewhere in the model.

💡 *please refer to **Host-Target Communication – Fundamentals in DL PEL-HIL Getting Started** to understand how the serial communication between host and target works, and what parse blocks is used for.*




In this model, the data received from the host include **Relay_host**, **Discharge_host**, **Enable_host**, **Reset**, **Calibrate**, D_**host and modulation.** All of them are stored in the data store memory with their data types specified.

**The second subsystem**, triggered by the ADC interrupt, performs data acquisition, converter control, protection, and data transmission to the host.

💡 *To enable periodic sampling, the modules in the DL Library are designed as follows: using a specific point of the carrier that generates the V1P switching pulse as the reference, the Start of Conversion (SOC) for analog data acquisition is triggered. When the data acquisition of VL1 (of DL RCPCORE) is completed, an interrupt event is generated. This process repeats continuously. Therefore, this subsystem operates at the PWM frequency of V1P.*

*Ensure V1 or V1P/N block exists if any analog data acquisition block is used.*

*Ensure VL1 block and the V1 or V1P/N block exist if the Dual-INT or ADCA1-INT block is used.*

In this subsystem, add the following blocks from DL library:

- **ctrl en**
  Automatically added and configured by the CreateNew command.
  This block determines whether the DL RCPCORE control outputs are enabled. It prevents the outputs from being undefined during target model compilation and deployment. The outputs will be enabled only after deployment, so the input of this block is often set to a fixed Boolean value of 1.

- **VL1**
  Automatically added and configured by the CreateNew command.
  VL1 measures high voltage from channel VL1 of the DL RCPCORE. This block triggers ADCINT1 interrupt and is essential for scheduling with ADC interrupt whether it will used or not.
- **V1**
  Automatically added and configured by the CreateNew command.
  The block controls the 1st IGBT leg of the inverter of DL 2106T06 from the Simulink model by duty cycle ranging from 0 to 1.
  It enables the Start of Conversion (SOC) for all other analog measurement blocks. Therefore, it must be included whenever analog signal measurement is required.
  To keep leg 1 from being uncontrolled, V1 or V1P/N must be present with a valid duty cycle, even if the leg is not used in the topology.
- **V2**
  Automatically added and configured by the CreateNew command.
  The block controls the 2nd IGBT leg of the inverter of DL 2106T06 from the Simulink model by duty cycle ranging from 0 to 1.
  To keep leg 2 from being uncontrolled, V2 or V2P/N must be present with a valid duty cycle, even if the leg is not used in the topology.
- **V3**
  Automatically added and configured by the CreateNew command.
  The block controls the 3rd IGBT leg of the inverter of DL 2106T06 from the Simulink model by duty cycle ranging from 0 to 1.
  To keep leg 3 from being uncontrolled, V3 or V3P/N must be present with a valid duty cycle, even if the leg is not used in the topology.
- **Fault**
  This block reads whether the DL 2106T06 has a hardware fault.
- **Relay**
  This block controls the input relay of the DL 2106T06.
- **Vdc**
  This block acquires the DC link voltage from the DL 2106T06.
- **Iinv**
  This block acquires the DC link output current, i.e., the inverter's supply current.

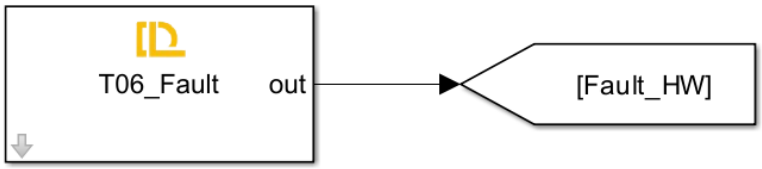
## Step 2: Target Model Fault Detection & Protection Logic

1. **Fault detection**

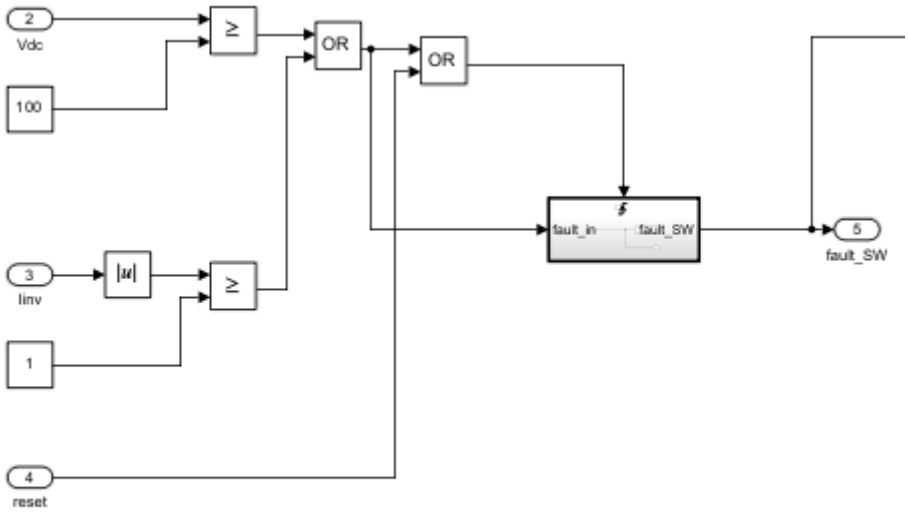
   Faults are classified into two types: hardware-reported DL 2106T06 faults and software-detected faults defined by the user for the application.

   **Fault_HW**

   As mentioned earlier, the Fault block monitors hardware faults in the DL 2106T06.




   To ensure safe operation, a fault is triggered when the DC link exceeds 100 V or the absolute inverter supply current exceeds 1 A. Once a fault occurs, it is latched — meaning the fault remains active even if the conditions return to normal. The fault can only be reset if the user presses the reset button and both the voltage and current fall below their respective thresholds.



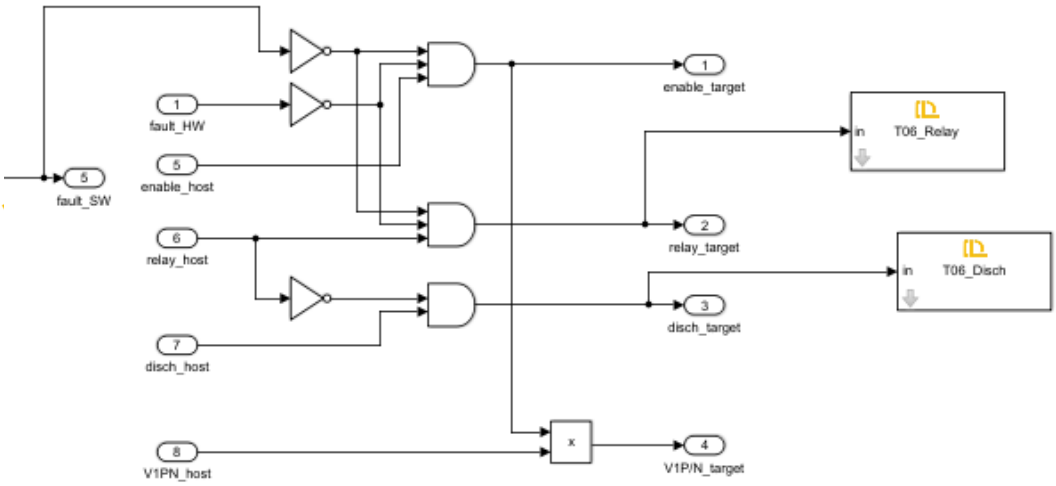

2. **Protection Logic**

   In the target model, several control constraints are implemented to ensure safe and reliable operation of the buck converter. These include:

   - The **enable_target** signal is activated only when:
     - enable_host = 1 **and**
     - No hardware or software faults are present.
   - The **Relay_target** signal (which allows connection of the AC source to the DC link) is set only when:

- relay_host = 1 **and**
- No hardware or software faults are present.

- The **discharge_target** signal (used to safely discharge the DC link capacitor) is activated only when:
  - Discharge_host = 1 **and**
  - Relay_target = 0 (ensuring no charging source is connected while discharging to avoid overload).
- The **D_target** signal is updated with the value from the host (D_host) **only if** enable_target = 1; otherwise, it is forced to zero.

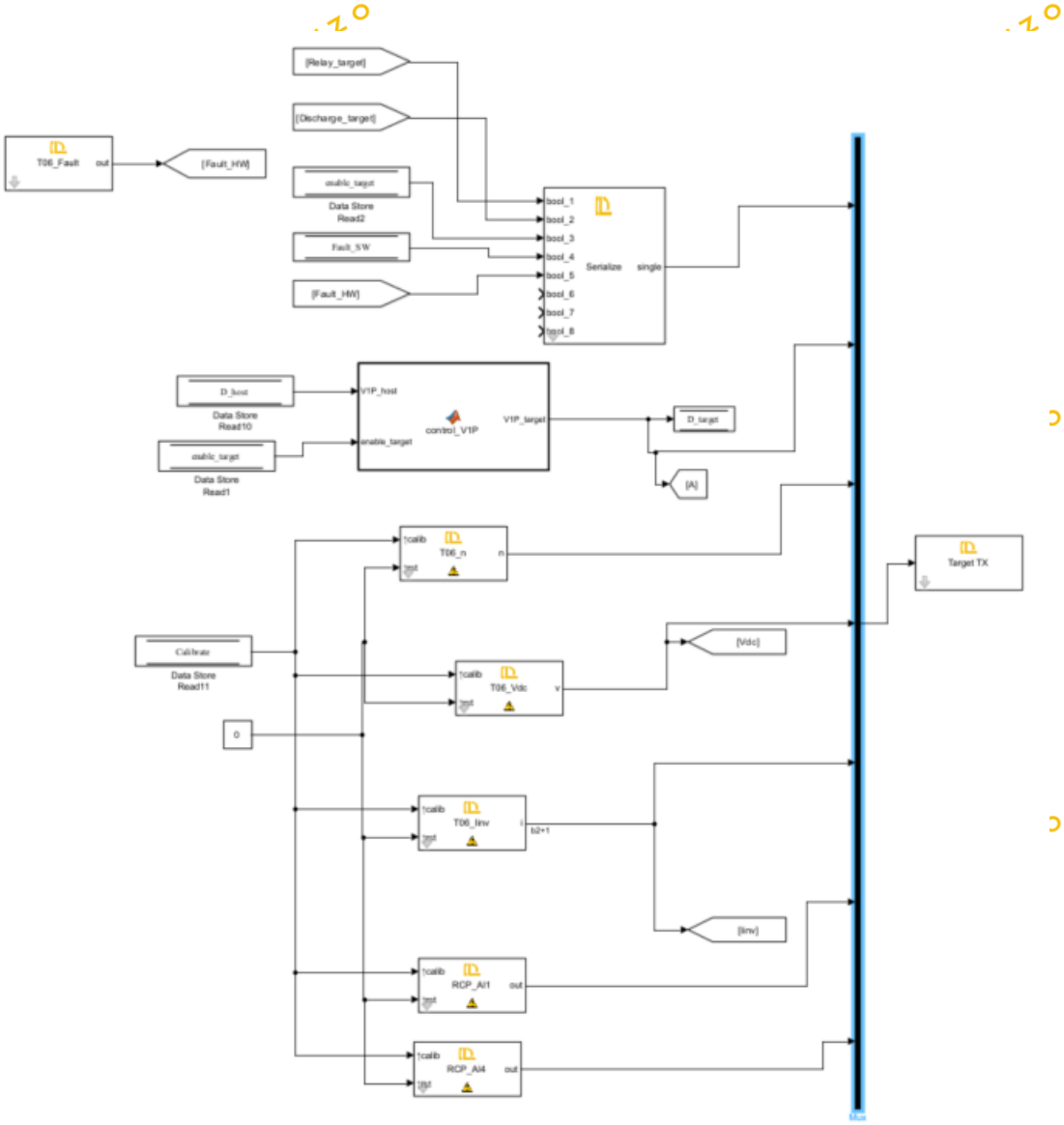



These constraints are essential for the following reasons:

- **Safety**: They prevent power flow or switching commands when a fault is detected, reducing the risk of hardware damage or electrical hazards.
- **Controlled Discharge**: They avoid discharging the capacitor while the relay is still connected to the power supply, which could cause dangerous short circuits.
- **Predictable behaviour**: Ensuring switching only occurs when explicitly allowed by both host and safety logic improves the reliability and traceability of system behaviour.

## Step 3: Target Model TX

To observe the target's real-time operation in the Host model, signals including relay, DC discharge, control enable, software and hardware faults, Vdc, Iinv, etc, are transmitted back to the host.

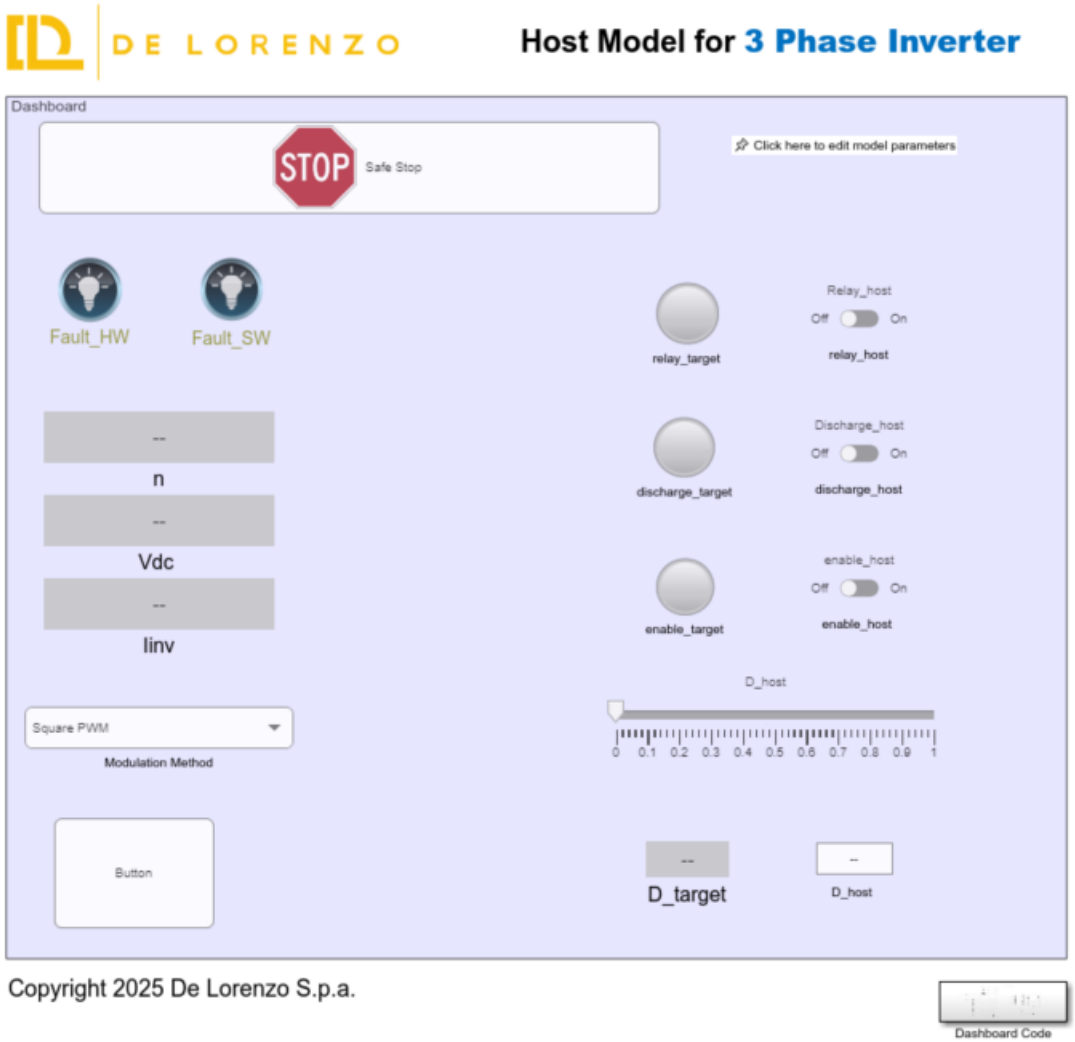



See tutorial 'Host-Target Communication' for details.

## Step 4: Target Model Build, Deploy and Start

Once deployed, the DL RCPCORE runs the target model continuously at startup, while certain parameters and commands rely on real-time host communication.

## Step 5: Host Model




The Host model consists of:

1. **Host–Target communication.**
   - RX blocks (to receive status, measurements, fault flags).
   - TX blocks (to send commands: Relay_host, Discharge_host, enable_host, Reset, D_host, etc.).
   - Proper framing and timing configuration.

2. **Dashboard**
    - Switches and knobs:
        - Relay command
        - Discharge command
        - Enable command.
        - Reset button.
        - Duty command (D_host slider).
    - Displays and scopes:
        - DC link voltage (Vdc)
        - Inverter current (Iinv)
        - Fault indicators
        - Duty values

For details on how to implement the Host side communication and dashboard, refer again to the **“Host–Target Communication”** tutorial.